Ratios and matching amounts

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Round 448 to the nearest hundred.

2. Round 4.72 to the nearest whole number.

3. Use 20 × 5 to estimate 19.8 × 5.1.

4. Use 60 ÷ 3 to estimate 59.4 ÷ 3.1.

5. Is 6 : 10 equivalent to 9 : 15?

6. Ava changes 3 : 7 to 5 : 9. Is it the same ratio?

7. Simplify the ratio 16 : 28 by dividing both counts by 4.

8. Write a story for $\frac{1}{6} \div 3$ and solve it.

9. Find $\frac{1}{2} + \frac{1}{3}$.

10. Find $\frac{2}{3} + \frac{1}{4}$.

11. $\frac{4}{3} - \frac{3}{4}$

12. $\frac{3}{2} - \frac{3}{4}$

13. A team has 4 runners for every 3 walkers. There are 18 walkers. How many runners?

14. Leo says 2 yellow : 3 blue means $\frac{2}{3}$ of all the paint is yellow. Repair his statement.

15. Red: blue is 3 : 6. Complete (3, 6), (9,__).

16. A mix has 16 raisins and 16 nuts. Divide both counts by 4; write the new ratio.

17. A recipe uses 10 cups oats for 12 cups seeds. How many cups seeds go with 15 cups oats?

18. Gold: silver beads are 2 : 3. There are 15 beads. Find each count.

CHALLENGE / Explain your plan and check the result

C1. A recipe uses flour: sugar 5 : 2. You have 20 cups flour and 10 cups sugar. Use all the flour. Find the sugar used, sugar left, and total ingredients in the batch.

C2. A row repeats 2 triangles then 3 squares. How many of each shape occur in the first 37 shapes? State the triangle: square ratio for those 37 shapes.